

Computational Linguistics 1 (2026)

Assignment 3: Phase 1

(Manual and LLM-Assisted Annotation)

Deadline: March 24, 2026 (11:59 PM)

Objective

This phase focuses on the foundation of any NLP pipeline: **Data**. You will create a high-quality manually annotated test set and a larger LLM-augmented training set for POS tagging, Chunking, and NER.

Task 1: Manual Annotation (20 Marks)

This task requires manual annotation of 100 sentences in total. You must perform these **without the use of any automated tools**.

Task 1a: UD-based Annotation (10 Marks)

Select **50 sentences** (at least 350 tokens) from your previous datasets.

- **POS Tags:** Use the **Universal Dependencies (UD)** tagset.
- **Reference:** UD POS Tagset Documentation.
- **Chunking & NER:** Annotate using standard IOB format.

Task 1b: BIS-based Annotation (10 Marks)

Select **50 sentences** (at least 350 tokens) from any **raw** Indian language dataset.

- **Tagset:** Use the **BIS (Bureau of Indian Standards)** tagset.
- **Reference:** Bhashini BIS Guidelines.
- **Chunking & NER:** Annotate using standard IOB format.

Task 2: Local LLM-based Annotation (10 Marks)

Expand your training data by generating **200 sentences for English** and **200 sentences for your chosen Indian Language** (400 total).

Technical Requirement: You must use a **local LLM** (e.g., Gemma, Llama 3) via a local API (Ollama, vLLM, or LM Studio).

Zero-shot vs. Few-shot Annotation

You should experiment with both:

- **Zero-shot:** Prompt the LLM with instructions only.
- **Few-shot:** Provide 3–5 examples in the prompt.

Prompt the LLM to annotate sentences without examples.

Example prompt:

```
Annotate the following sentence with POS, Chunk, and NER tags
in CoNLL format.
```

```
Sentence:
```

```
Ram went to Delhi yesterday.
```

```
Example Few-shot Prompt:
```

```
Ram PROPN B-NP B-PER
```

```
went VERB B-VP 0
```

```
to ADP B-PP 0
```

```
Delhi PROPN B-NP B-LOC
```

```
Now annotate:
```

```
The Prime Minister visited Hyderabad.
```

Python Snippet (Ollama API)

```
import requests
import json

def get_llm_annotation(prompt_text):
    url = "http://localhost:11434/api/generate"
    payload = {
        "model": "gemma2", # or llama3
        "prompt": prompt_text,
        "stream": False
    }
    response = requests.post(url, json=payload)
    return response.json()['response']
```

Submission Instructions

Submit a ZIP file named <Roll_number>.zip containing:

- `manual_dataset.conll` (The 100 manually annotated sentences)
- `llm_dataset.conll` (The 400 LLM-annotated sentences)
- `phase1_report.pdf` (Prompts used and post-editing details)